

DUBLIN AP, Ireland

WMO: 039690

Lat: 53.433N

Lon: 6.250W

Elev: 74

StdP: 100.44

Time Zone: 1.00 (IRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	-1.2	-4.4	2.6	-0.7	-3.0	3.0	0.3	16.4	9.4	15.2	9.0	3.3	260	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	22.1	17.0	20.6	16.3	19.4	15.7	17.9	20.7	17.1	19.6	16.3	18.7	5.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	11.9	19.2	15.9	11.4	18.5	15.2	10.9	17.7	50.5	20.8	48.1	19.7	46.0	18.7	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.4	11.9	10.6	-5.5	24.4	1.8	1.5	-6.7	25.5	-7.8	26.3	-8.7	27.2	-10.0	28.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-5.6	19.4	1.6	0.9	-6.8	20.0	-7.7	20.6	-8.6	21.1	-9.8	21.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	5.3	5.4	6.4	8.1	10.6	13.3	15.2	15.0	13.1	10.5	7.4	5.7
	DBStd	4.45	2.88	2.77	2.73	2.56	2.42	2.24	2.05	2.00	2.23	2.65	2.93	3.25
	HDD10.0	739	147	132	116	68	22	2	0	0	2	27	87	137
	HDD18.3	3164	404	363	371	307	239	151	99	106	158	244	329	393
	CDD10.0	619	1	2	3	10	41	101	161	154	94	41	8	3
	CDD18.3	4	0	0	0	0	0	1	2	2	1	0	0	0
	CDH23.3	12	0	0	0	0	0	3	6	3	1	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.3	6.3	6.0	5.8	5.2	5.1	4.7	4.5	4.7	4.9	5.4	5.6	5.9
Precipitation	PrecAvg	747	67	50	53	54	57	59	56	69	63	76	74	73
	PrecMax	1032	119	128	107	147	150	148	165	167	145	181	187	217
	PrecMin	556	14	5	5	8	4	5	9	7	3	11	21	15
	PrecStd	100	29	28	25	27	30	34	31	35	34	41	40	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	13.4	14.9	17.4	20.4	23.2	24.1	23.4	21.6	17.9	15.2	14.0
		MCWB	11.4	11.2	10.9	12.9	15.2	16.4	18.0	18.5	17.1	15.2	13.7	12.4
	2%	DB	11.8	12.0	13.1	15.4	18.2	20.7	22.2	21.2	19.5	16.5	13.8	12.7
		MCWB	10.4	10.1	10.4	11.6	14.0	15.9	17.2	17.1	16.2	14.3	12.5	11.2
	5%	DB	10.9	11.0	12.0	13.8	16.6	19.2	20.7	19.9	18.1	15.6	12.9	11.6
		MCWB	9.6	9.5	9.8	10.6	12.9	15.0	16.4	16.2	15.1	13.6	11.5	10.3
	10%	DB	9.9	9.9	10.8	12.6	15.1	17.9	19.3	18.8	17.1	14.7	11.9	10.6
		MCWB	8.7	8.4	8.9	9.8	11.9	14.1	15.7	15.6	14.3	13.0	10.7	9.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.7	12.0	13.5	16.0	17.8	18.9	19.1	17.9	15.8	14.0	12.6
		MCDB	12.8	12.9	13.4	16.2	19.6	21.3	22.9	22.4	20.5	17.0	14.9	13.6
	2%	WB	10.7	10.6	10.9	12.1	14.5	16.4	17.9	17.8	16.7	14.8	12.8	11.4
		MCDB	11.6	11.7	12.5	14.7	17.4	19.6	21.0	20.3	18.8	16.1	13.6	12.4
	5%	WB	9.8	9.6	10.0	11.0	13.4	15.5	17.0	17.0	15.8	14.0	11.7	10.5
		MCDB	10.8	10.8	11.7	13.3	15.9	18.5	19.9	19.0	17.6	15.3	12.6	11.4
	10%	WB	8.8	8.6	9.1	10.1	12.4	14.6	16.2	16.2	14.9	13.2	10.8	9.5
		MCDB	9.7	9.8	10.7	12.1	14.7	17.4	18.7	18.2	16.5	14.4	11.8	10.5
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.8	6.7	7.4	7.7	7.9	7.6	7.3	7.1	6.2	5.4	5.2
		MCDBR	6.2	6.2	7.4	8.8	9.6	9.7	9.3	8.3	8.3	6.5	5.5	5.9
		MCWBR	5.7	5.3	5.3	5.7	6.2	5.9	5.5	5.0	5.5	5.0	5.0	5.5
	5% WB	MCDBR	6.0	6.0	6.7	8.2	8.7	9.0	8.4	7.6	7.4	5.9	5.4	5.9
		MCWBR	5.8	5.4	5.3	5.7	6.0	5.9	5.4	5.1	5.6	5.0	5.1	5.6
Clear-Sky Solar Irradiance	taub	0.336	0.346	0.374	0.389	0.396	0.385	0.388	0.369	0.362	0.349	0.328	0.324	
	taud	2.260	2.330	2.280	2.271	2.300	2.363	2.360	2.426	2.440	2.419	2.334	2.235	
	Ebn at Noon	611	745	801	839	853	865	855	853	812	736	623	542	
	Edh at Noon	64	81	104	119	121	115	114	101	88	73	59	54	
All-Sky Solar Radiation	RadAvg	0.68	1.39	2.51	3.87	4.92	5.13	4.65	3.91	2.89	1.68	0.86	0.52	
	RadStd	0.06	0.12	0.24	0.36	0.36	0.51	0.41	0.34	0.16	0.14	0.09	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page